Multimodal Large Language Models versus Task-Specific Models for Ordinal Student Attention Recognition

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Abstract. Multimodal large language models (MLLMs) have become very good at many visual tasks, which has led many to believe that a single pre-trained model can replace a model built for one specific task. We test this idea directly for in-classroom student attention recognition, a task with three features that make transfer hard: the answer is an ordered rank, the visual signal is subtle and hard to rate, and labelled data are scarce. Using the DIPSER dataset and an identical evaluation protocol established in a prior benchmark on DIPSER, we compare four settings: (1) a task-specific Tchebichef-coefficient pipeline (Task-specific baseline) that set the first published benchmark on DIPSER; (2) Zero-shot prompting of a 7-billion-parameter MLLM; (3) Few-shot prompting with labelled examples per class; and (4) parameter-efficient LoRA fine-tuning of the same MLLM. Our main metric is quadratic weighted kappa (QWK), which penalises ordinal misalignment; we also report mean absolute error, macro-F1, and accuracy. Results show a QWK of 0.137 for zero-shot, 0.020 for few-shot (degenerate: the model copies its last example), and 0.282 ± 0.236 for LoRA fine-tuning (five-fold), against 0.230 ± 0.154 for the task-specific baseline under the same corrected protocol. The two adapted settings are therefore roughly equal, while unadapted prompting does not work: zero-shot only follows the class frequencies, and few-shot collapses onto a predictor that points the opposite way. A 7B MLLM can thus handle fine-grained ordinal classroom attention once adapted, requiring a few hours of rank-8 LoRA on a single GPU; the key distinction is *adapted* vs. unadapted, not the model family. These results set up a multi-condition benchmark on DIPSER and help decide between foundation models and task-specific models for educational affect recognition.

Keywords: Multimodal large language models · Student attention recognition · Foundation models · Parameter-efficient fine-tuning · LoRA · Zero-shot evaluation · Ordinal classification · DIPSER dataset · Quadratic weighted kappa.

1 Introduction

A teacher watching a classroom cannot see every student at once. Automatic attention recognition would fill that gap: a camera on each desk detects when a student drifts into distraction, so the teacher can intervene then and there, and adaptive tutoring systems can adapt the lesson in real time [12]. The same face-based analytics already support attendance tracking [5], and extending them to attention is the natural next step.

The difficulty is not engineering, it is the labels. Attention is a continuum, and human raters disagree about it: on a 1–5 scale the experts themselves rarely agree better than a quadratic-weighted $\kappa \approx 0.1$. This is a much harder setting than the clean image–label tasks on which modern AI models are usually judged. And the label distribution is highly skewed; in our corrected slice the "high" class appears in only 0.5% of the labelled frames.

Meanwhile, multimodal large language models (MLLMs) such as GPT-4o, LLaVA, and Qwen2-VL have shown that one big pre-trained model can answer many visual questions at once, with no task-specific training. It is tempting to assume the same model can simply read a face in the classroom and rate attention correctly. Three features of this task make that hope fragile:

(i) **The answer is an order, not a category.** Attention lives on a 1–5 scale (reduced to low, medium, or high for statistical power), and MLLMs are not trained to respect ordered ranks.
(ii) **Raters disagree.** The visual gap between attention 2 and attention 3 is small and genuinely ambiguous, even for expert raters.
(iii) **Labelled data are scarce.** The benchmark slice has 20 subjects, far from the abundant data on which MLLM transfer is demonstrated, and the only published exploration of foundation models on it is the dataset authors' own few-shot study [10].

We turn this fragile hope into a controlled measurement. Using the DIPSER dataset [9] and the same GroupKFold-by-subject protocol as the prior benchmark on DIPSER, we compare four settings: zero-shot prompting (no training), few-shot prompting (a few labelled examples in the prompt), parameter-efficient LoRA fine-tuning, and the task-specific Tchebichef baseline from the prior benchmark. Every setting is scored with the same ordinal metrics (QWK, MAE, macro-F1), so the comparison is fair and reproducible.

Contributions.

1. The first *controlled, multi-setting* comparison of MLLMs against a task-specific pipeline on DIPSER (zero-shot, few-shot, LoRA) under one pre-specified ordinal protocol, complementing the dataset authors' few-shot exploration [10].
2. Evidence, under a corrected identical protocol, that the *adapted* MLLM and the task-specific baseline perform equally (QWK 0.282 ± 0.236 for LoRA vs 0.230 ± 0.154), while unadapted prompting fails (0.137 zero-shot, 0.020 degenerate few-shot); the useful difference is *adapted* vs. unadapted.

3. An explanation of why unadapted models fail: zero-shot only follows the class frequencies, few-shot collapses onto a constant *anti*-correlated predictor (the model copies its last example), no setting recovers the rare "high" class, and this follows from the mismatch between MLLM pretraining and ordered rank prediction.

2 Related Work

Student engagement and attention recognition. Automatic recognition of student engagement has been studied through facial action units [15], head pose [11], and by combining video and physiological signals [1]. The usual targets are binary (engaged/disengaged) or coarse 3-class labels. Ordered attention under a multi-rater protocol is studied in prior work on DIPSER, and face–wearable fusion is reported on the same dataset [8]. Closest to our question, the DIPSER authors explore few-shot VLM/MLLM attention prediction for STEM education [10] and show it is possible; what they do not provide is a controlled four-setting comparison (zero-shot, few-shot, LoRA, task-specific) under one pre-specified ordinal protocol.

Multimodal large language models. Vision-language models pre-trained with contrastive objectives (CLIP [13]) made zero-shot visual classification possible. Instruction-tuned MLLMs such as LLaVA [7], InstructBLIP [3], and Qwen2-VL [14] then followed a generate-then-parse approach, with strong zero- and few-shot performance on natural-image benchmarks (VQA, COCO captioning, OCR). Performance drops sharply on specialised domains (medical imaging, remote sensing, affective computing) where the target signal is rare or absent in the web-scale pretraining data.

Parameter-efficient fine-tuning and ordinal classification. Low-rank adaptation (LoRA [6]) adds small trainable rank-r matrices to the attention projections of a frozen backbone, training only a small fraction of the parameters and giving competitive results on domain-shifted VLM datasets with a few thousand labelled samples. Whether this holds for ordered tasks with fewer than 20 labelled subjects is our central question. Ordinal regression predicts a ranked output: CORAL [2] and ordinal cross-entropy [4] are standard, but both need task-specific heads that instruction-tuned MLLMs do not have. Prompting an MLLM to say a rank word ("low"/"medium"/"high") skips the ordinal loss entirely, and whether that text output is ordinally consistent is an open empirical question.

3 Experimental Setup

3.1 Dataset

We use the same slice of DIPSER [9] as the prior benchmark: `group_01` experiments 01 and 05, 20 subjects, 99,892 usable frames at about 10 fps from individual face cameras. Labels are the majority vote across five raters (four experts plus student self-report), mapped to three ordered classes: *low* (1–2), *medium* (3),

and *high* (4–5). The evaluated slice contains 1,248 face frames from 20 student–experiment bundles; its clean-label distribution is low 48.3%, medium 51.2%, high 0.5% (dist $[603, 639, 6]$): the "high" class is so rare that it cannot be recovered with certainty, which (as in prior work) makes it the hardest class and limits the achievable QWK. (An earlier published slice reported $[587, 605, 56]$; the clean label pipeline could not reproduce that distribution, so the corrected $[603, 639, 6]$ slice is used throughout.) Inter-rater quadratic-weighted κ ranges from -0.10 to $+0.22$, confirming that the labelling task is inherently ambiguous. Full dataset details are in prior work on DIPSER.

3.2 Evaluation protocol

All four settings share the same evaluation protocol to remove confounders:

- **Splits:** five-fold `GroupKFold` by subject (a subject's frames never appear in both train and validation), identical to the prior benchmark.
- **Metrics:** quadratic weighted kappa (QWK, primary), mean absolute error (MAE), macro-F1, accuracy, and Matthews correlation coefficient (MCC). QWK punishes ordinal misalignment quadratically and is the most informative metric for this task.
- **Reporting:** mean $\pm$ std over the five folds.
- **Input resolution:** 224×224 RGB face crops for all MLLM settings (the MLLM's native size); the task-specific baseline used 64×64 grayscale, which is noted where relevant.

3.3 Hardware and software

MLLM inference and LoRA fine-tuning run on Kaggle GPU T4 ($\times 2$, 16 GB VRAM each). The base model is **Qwen2-VL-7B-Instruct** [14] loaded in 4-bit NF4 quantisation (bitsandbytes), which lowers peak VRAM to about 5 GB for inference and 12 GB during LoRA training. Software: Python 3.11, Transformers 4.x, PEFT 0.x, bitsandbytes 0.x.

4 Experimental Conditions

4.1 Condition 1: Task-specific baseline (prior benchmark)

The best configuration from the prior benchmark: DenseNet (growth rate 16, compression 0.5) with HP_B hyperparameters (GELU, lr 6.31×10^{-3}), CORAL ordinal loss, and Tchebichef-coefficient input representation. This model was trained from scratch on 64×64 grayscale face crops; architecture and training details are in prior work on DIPSER. We import the per-fold JSON results directly without re-running.

4.2 Condition 2: Zero-shot MLLM

The model receives a single system prompt defining the task and one user turn containing the face image:

> *"You are an expert educational psychologist. Rate this student's attention level from their face image. Respond with exactly one word: low, medium, or high."*

Table 1. Comparison of all conditions on DIPSER attention (3-class ordinal). QWK is the primary metric. ↑ = higher is better; ↓ = lower is better. Best per column in **bold**. LoRA and the task-specific baseline are five-fold subject-disjoint means ± std; zero-/few-shot are evaluated over the full 1,248-frame slice (few-shot collapsed: the model echoes its last in-context demonstration).

Condition	QWK ↑	MAE ↓	Macro-F1 ↑	Acc ↑
LoRA (Qwen2-VL-7B, 5-fold)	$\mathbf{0.282 \pm 0.236}$	**0.345**	**0.428**	**0.659**
Task-specific baseline (DenseNet·CORAL)	0.230 ± 0.154	0.371	0.406	0.632
Zero-shot (Qwen2-VL-7B)	0.137	0.454	0.353	0.556
Few-shot (Qwen2-VL-7B, degenerate)	0.020	1.368	0.068	0.060

No labelled training data is used. The model's response is parsed to $\{0, 1, 2\}$; when the response does not contain any class word, it falls back to the majority class (medium).

4.3 Condition 3: Few-shot MLLM

Identical to zero-shot except that five (image, label) example pairs per class (15 examples total) are added as demonstrated assistant turns before the query image. Examples are drawn uniformly at random from the training fold, so there is no leakage from the validation fold. The examples always contain 5 per class, independent of the training fold's natural class distribution.

4.4 Condition 4: LoRA fine-tuning

A rank-8 LoRA adapter [6] is attached to the query and value projections of the MLLM's attention layers. The base model weights stay frozen; only the adapter parameters are trained (2.52M trainable parameters out of 4.69B counted by the training harness, i.e. 0.05%). Training uses the same zero-shot prompt format with the ground-truth label added as the completion; the causal language modelling loss is masked over the prompt tokens so only the one-word answer is supervised. Optimiser: AdamW, lr 2×10^{-4}, batch size 4, two epochs. A fresh adapter is initialised per fold; there is no adapter sharing across folds.

5 Results

5.1 Main comparison

Table 1 reports QWK, MAE, macro-F1, and accuracy for all four settings. LoRA and the task-specific baseline are five-fold subject-disjoint means; the two prompting settings are evaluated over the full 1,248-frame slice under the identical clean-label protocol.

The result is more nuanced than the earlier single-fold run suggested. LoRA reaches the best QWK (0.282 ± 0.236) and the task-specific pipeline (Task-specific baseline, 0.230 ± 0.154) is statistically indistinguishable from it given the fold-wise standard deviations. Zero-shot reaches a modest 0.137; few-shot collapses to 0.020 QWK and 0.060 accuracy, the model copying its last in-context demonstration rather than classifying. The message is therefore not about model size: *adapted* vs. unadapted is what matters, and adaptation closes the gap to the dedicated pipeline.

5.2 Class-level breakdown

On the corrected slice (true distribution $[603, 639, 6]$ for low/medium/high), the predicted distributions show the failure modes directly. Zero-shot predicts $[953, 276, 19]$, loosely following the class frequencies: QWK 0.137, accuracy 0.556. Few-shot predicts $[114, 0, 1134]$, reversing the true proportions: the model answers "high" for 91% of frames because the last in-context demonstration is a high-attention example. LoRA separates the two majority classes (pooled low/medium recall 0.715/0.623; confusion $[[431, 172], [241, 398]]$), raising QWK to 0.282 (five-fold mean; 0.320 pooled). No setting recovers the rare "high" class ($\approx 0.5\%$ of the slice), the same class collapse seen in the prior benchmark.

5.3 Ordinal consistency

The effect of the ordinal weighting is clearest in few-shot: its MAE of 1.368 is the largest of any setting because it inverts the label order, whereas zero-shot, by concentrating on the majority classes, keeps MAE at 0.454. The two adapted settings earn the lowest mean absolute errors (0.345 for LoRA, 0.371 for Task-specific baseline). QWK, because it punishes distance to the true ordinal label, is what separates informative from uninformative predictions; this is exactly why it is the right primary criterion for noisy ordinal labels.

All MLLM settings returned a parseable class word for essentially every frame, so the failure of zero-shot and few-shot is *semantic*, not a formatting problem: the model answers fluently but either follows only the class frequencies or copies its last demonstration.

6 Discussion

Why unadapted MLLMs fail, and what LoRA fixes. The low QWK of the unadapted settings follows from a mismatch between MLLM pretraining and the target task. Web-scale image–text data contains almost no fine-grained ordinal classroom affect, so the model's prior over "low"/"medium"/"high" is shaped by unrelated contexts and is not aligned with the rater-calibrated ordinal scale in DIPSER. A few in-context examples point the model at the right vocabulary but cannot teach the calibration of the 5-rater labelling process, so zero-shot only follows the class frequencies and few-shot degenerates into a copy of its last demonstration (answering "high" for 91% of frames). LoRA fixes this: once exposed to the training fold's class distribution, it breaks the collapse across the five folds and lifts QWK to 0.282 ± 0.236 while training only 0.05% of the parameters on a 4-bit-quantised base. That puts it statistically on par with the purpose-built CORAL model's 0.230 ± 0.154 on the ordinal metric, while still never predicting "high": the causal-LM loss punishes an off-by-one and an off-by-two error identically, so the rare class remains unrecoverable without an explicit ordinal loss. Closing this architectural gap goes beyond standard PEFT: it would require replacing the generation head with a CORAL-style ordinal head, the natural next step toward minority-class recovery.

Practical guidance. **A cheap task-specific model and a cheaply adapted 7B MLLM sit at the same point** on fine-grained ordinal classroom attention:

the corrected five-fold numbers make the two statistically indistinguishable, and LoRA needs only one GPU for a few hours. **Prefer the lightweight model** when deployment cost, power, or latency dominate, or when only the majority classes matter. **Consider a foundation model** only when labelled data are too scarce to train a dedicated network, when broad cross-dataset coverage is needed, or when a mid-size GPU is cheap; in all such cases, always adapt, since off-the-shelf prompting collapses. Never deploy an unadapted MLLM to monitor attention: its fluent assessment is systematically misleading.

Limitations. We evaluate a single MLLM architecture (Qwen2-VL-7B) and a single dataset (DIPSER attention slice); larger models (13B, 72B), other VLMs (LLaVA-1.6, InternVL2), and other engagement datasets may behave differently. The 4-bit quantisation adds some degradation relative to full precision; we report quantised results throughout for reproducibility on free-tier hardware. The high class is only 0.5% of the corrected slice, so its recovery is barely testable here. Future work should explore ordinal-head fine-tuning and multi-modal LoRA that incorporates sensor data alongside face images.

7 Conclusion

We ran a controlled four-setting comparison between multimodal large language models and a task-specific lightweight pipeline for ordinal student attention recognition on DIPSER, under a shared corrected protocol with QWK as the primary metric. Unadapted prompting of Qwen2-VL-7B does not transfer: zero-shot only follows the class frequencies (QWK 0.137 vs 0.230 ± 0.154 for the task-specific baseline) and few-shot degenerates into a copy of its last demonstration (0.020). LoRA decides the outcome: a rank-8 adapter lifts QWK to 0.282 ± 0.236 by training only 0.05% of the parameters in a few GPU-hours, statistically indistinguishable from the purpose-built CORAL baseline, but still never recovers the rare "high" class. An *adapted* 7B MLLM therefore matches a compact task-specific network on fine-grained ordinal classroom attention, while an unadapted one is unusable; the failure modes of unadapted use (small labelled data, ordinal targets, no ordinal loss) point to the fixes worth pursuing (ordinal-head PEFT, larger-scale data collection).

Reproducibility and ethics. All evaluation code, prompts, adapter configurations, and per-fold results are released in a public repository (link and archival DOI on acceptance). DIPSER was collected by its original authors under their own institutional approvals; this study uses only the publicly released data, with no new human-subjects collection and no re-identification. The negative result is itself ethically relevant: an unadapted foundation model would produce systematically misleading classroom assessments while appearing fluent and confident.

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